

RPS AI Bot Tournament Report

Five current shortcut bots and two historical versions, tested in five repeat runs.

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7 bot versions Five current, two historical	105,000 games compared Twenty-one opponent pairings	5 random seeds Seeds 42, 43, 44, 45, and 46	1st Aggressive Most game wins overall
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RPS AI offers five ways to choose a move: Regular, Aggressive, Short-Term Memory, Random, and Pattern Detection. This report compares them in simulated rock-paper-scissors games and includes two older versions to show how the recent changes affected their behavior.

Aggressive wins the most games overall: 16,439 out of 30,000, or 54.8%. Current Pattern Detection is second with 16,203 wins (54.0%). Pattern Detection has fewer losses and the largest wins-minus-losses margin, so it leads when judged by that measure. The standings and charts on the next pages show wins, losses, and ties together.

The five current bots use different approaches: Regular and Aggressive track frequently played moves, Short-Term Memory focuses on recent moves, Random makes weighted random choices, and Pattern Detection looks for repeated sequences. The older versions are labeled Pattern Detection (pre-v7.2) and Short-Term Memory (pre-v7.1).

Each bot plays the other six under the same test rules. We repeat the tournament five times, changing the random seed each time. A seed is a number that makes a random sequence repeatable. This helps show whether a result holds up across different runs. These tests compare simulated bots; they do not predict how often a bot will beat a person.

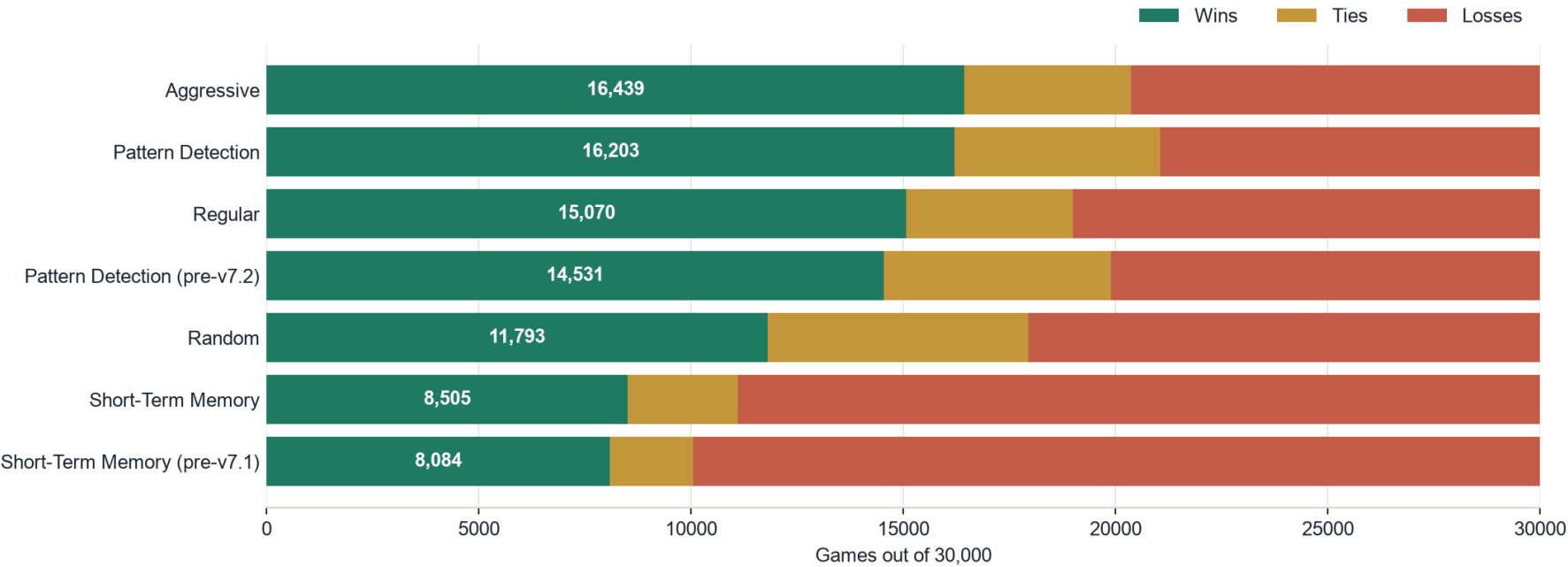
How to read the results. A game contains five rounds. The bot that wins more rounds wins the game; equal scores mean a tied game. W, L, and T mean wins, losses, and ties. Win % is the share of all games won, including ties in the total. A margin of +100 means 100 more wins than losses. Later pattern charts count individual rounds rather than whole games.

Results and charts: pages 2-6. Bot rules and version changes: pages 7-8. Detailed pattern analysis and test method: pages 9-11. Every bot matchup: pages 12-15. PDF bookmarks provide direct navigation.

Combined standings

Aggressive finishes first by total game wins. Each bot played 30,000 games. Win % is wins divided by all games played; margin is wins minus losses. All results combine the five runs.

Place	Bot	Game wins	Game losses	Game ties	Win %	Margin
1	Aggressive	16,439	9,641	3,920	54.8%	+6,798
2	Pattern Detection	16,203	8,946	4,851	54.0%	+7,257
3	Regular	15,070	11,005	3,925	50.2%	+4,065
4	Pattern Detection (pre-v7.2)	14,531	10,108	5,361	48.4%	+4,423
5	Random	11,793	12,063	6,144	39.3%	-270
6	Short-Term Memory	8,505	18,903	2,592	28.4%	-10,398
7	Short-Term Memory (pre-v7.1)	8,084	19,959	1,957	26.9%	-11,875



All twenty-one combined matchups

Each row combines all five runs. Wins, losses, and ties are shown from Bot A's side: 5,000 games and 25,000 rounds per row.

Bot A	Bot B	Game W	Game L	Game T	Margin	Round W	Round L	Round T
Aggressive	Pattern Detection	1,849	2,136	1,015	-287	8,869	9,590	6,541
Aggressive	Regular	2,423	1,905	672	+518	11,821	10,417	2,762
Aggressive	Pattern Detection (pre-v7.2)	1,912	2,106	982	-194	8,775	9,360	6,865
Aggressive	Random	1,971	1,966	1,063	+5	8,433	8,452	8,115
Aggressive	Short-Term Memory	4,060	779	161	+3,281	17,049	7,294	657
Aggressive	Short-Term Memory (pre-v7.1)	4,224	749	27	+3,475	17,435	7,474	91
Pattern Detection	Regular	2,160	1,861	979	+299	9,177	8,463	7,360
Pattern Detection	Pattern Detection (pre-v7.2)	2,027	1,963	1,010	+64	8,622	8,656	7,722
Pattern Detection	Random	2,007	1,934	1,059	+73	8,398	8,210	8,392
Pattern Detection	Short-Term Memory	3,794	753	453	+3,041	16,882	5,856	2,262
Pattern Detection	Short-Term Memory (pre-v7.1)	4,079	586	335	+3,493	18,714	4,993	1,293
Regular	Pattern Detection (pre-v7.2)	1,815	2,139	1,046	-324	8,389	9,090	7,521
Regular	Random	1,984	1,968	1,048	+16	8,531	8,385	8,084
Regular	Short-Term Memory	3,642	1,208	150	+2,434	15,667	8,749	584
Regular	Short-Term Memory (pre-v7.1)	3,863	1,107	30	+2,756	16,208	8,682	110
Pattern Detection (pre-v7.2)	Random	2,059	1,971	970	+88	8,490	8,273	8,237
Pattern Detection (pre-v7.2)	Short-Term Memory	2,824	1,376	800	+1,448	12,310	7,836	4,854
Pattern Detection (pre-v7.2)	Short-Term Memory (pre-v7.1)	3,440	1,007	553	+2,433	15,006	7,036	2,958
Random	Short-Term Memory	1,983	2,007	1,010	-24	8,392	8,472	8,136
Random	Short-Term Memory (pre-v7.1)	1,971	2,035	994	-64	8,475	8,629	7,896
Short-Term Memory	Short-Term Memory (pre-v7.1)	2,382	2,600	18	-218	12,288	12,635	77

Every matchup at a glance

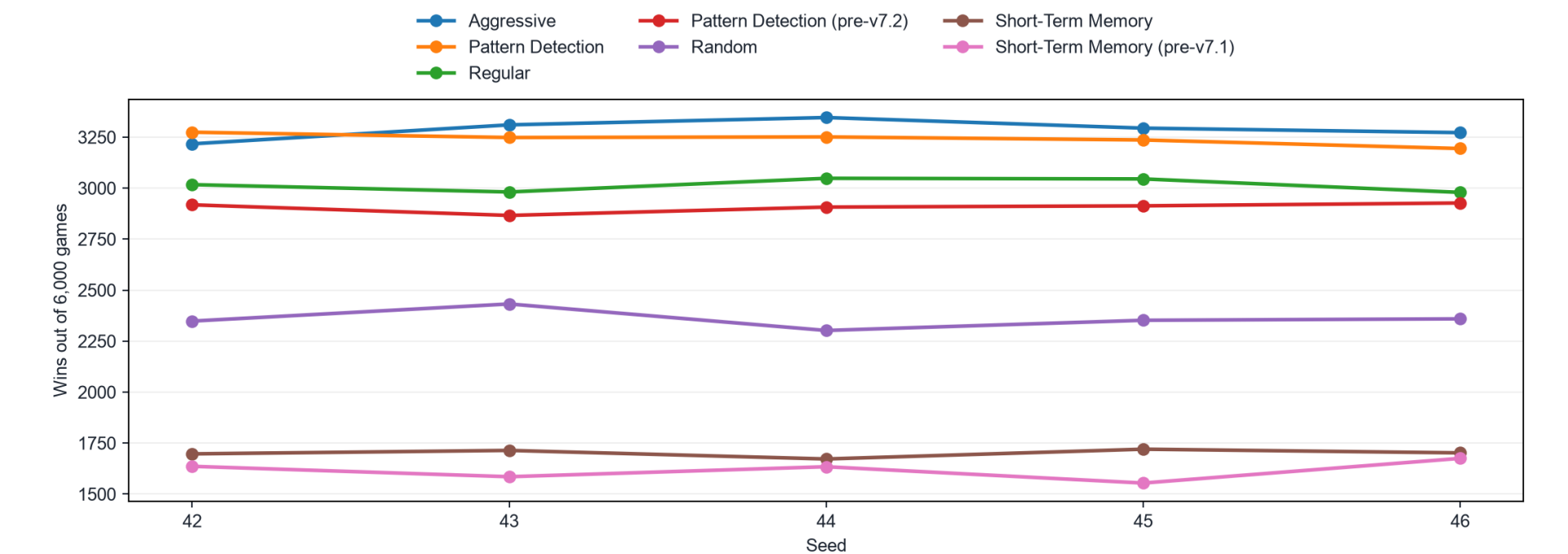
Choose a bot on the left and its opponent along the bottom. A positive number means the bot on the left won more games than it lost; a negative number means it lost more. Green shows an advantage and red a disadvantage.



How the five seeds compare

Each column is one repeat of the tournament, with a different random seed. The numbers show game wins, with the finishing place in parentheses. Every bot plays 6,000 games in each run.

Bot	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46	Combined W
Aggressive	3,217 (2)	3,310 (1)	3,346 (1)	3,294 (1)	3,272 (1)	16,439
Pattern Detection	3,274 (1)	3,248 (2)	3,251 (2)	3,236 (2)	3,194 (2)	16,203
Regular	3,017 (3)	2,981 (3)	3,048 (3)	3,045 (3)	2,979 (3)	15,070
Pattern Detection (pre-v7.2)	2,918 (4)	2,866 (4)	2,907 (4)	2,913 (4)	2,927 (4)	14,531
Random	2,348 (5)	2,432 (5)	2,302 (5)	2,352 (5)	2,359 (5)	11,793
Short-Term Memory	1,697 (6)	1,714 (6)	1,672 (6)	1,720 (6)	1,702 (6)	8,505
Short-Term Memory (pre-v7.1)	1,636 (7)	1,585 (7)	1,634 (7)	1,554 (7)	1,675 (7)	8,084

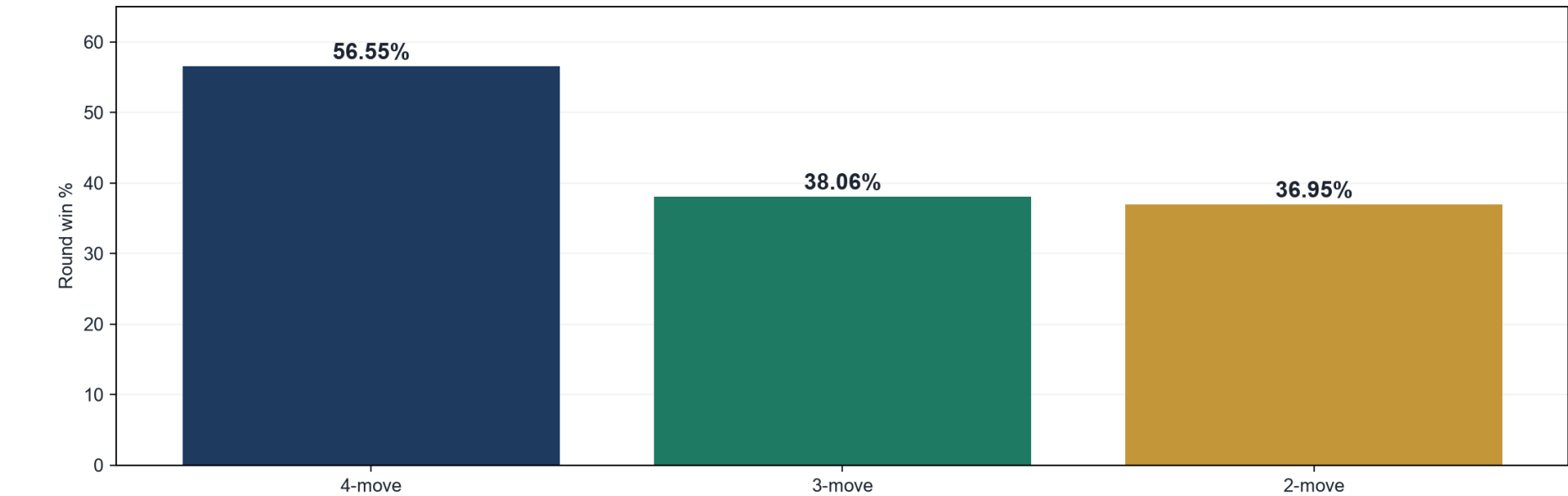


The rules stay the same in every seed. Each seed gives every bot the same number of games; combined rank is based on total wins across all five runs.

How successful were the selected patterns?

This section looks more closely at Pattern Detection. Each row groups rounds by the pattern length that actually chose its move. Regular fallback means no pattern matched; Regular warmup means the pattern history was still filling.

Choice source	Rounds	Wins	Losses	Ties	Round win %	Net score %
4-move	79,256	44,816	21,432	13,008	56.5%	+29.50%
3-move	44,104	16,787	14,644	12,673	38.1%	+4.86%
2-move	23,672	8,747	8,010	6,915	37.0%	+3.11%
Regular fallback	2,518	896	827	795	35.6%	+2.74%
Regular warmup	450	137	134	179	30.4%	+0.67%
ALL	150,000	71,383	45,047	33,570	47.6%	+17.56%



The bot won most often when a 4-move pattern chose its move. But it only tried 3 moves when 4 failed, and 2 when both failed, so the three groups faced different situations. This chart alone cannot prove that 4 moves is always better. Round win % counts all rounds, including ties. Net score % is wins minus losses, divided by all rounds.

How the bots learn and choose

To counter a move means to play the move that beats it: paper beats rock, scissors beats paper, and rock beats scissors. Where a bot keeps separate records by round position, it has one record for round 1, another for round 2, and so on through round 5.

Regular remembers the opponent moves from rounds it tied or lost. It plays the counter to the most frequent recorded move. If several moves are equally frequent, it chooses among their counters at random. With no recorded moves, it chooses randomly.

Aggressive works like Regular, but gives a loss twice as much weight as a tie. This makes it respond more strongly to moves that have beaten it.

Short-Term Memory uses the same basic choice as Regular, but keeps only the ten most recent opponent moves from tied or lost rounds in each round position. It leaves winning rounds out of its memory.

Random chooses at random, with some moves more likely than others. It combines the counts used by Regular with a second record that starts with one count for each move and adds a count when the bot wins. It samples an opponent move using those combined counts, then plays the counter.

Pattern Detection remembers the last fifteen opponent moves across all round positions. It looks for earlier sequences matching the most recent moves, uses what followed those sequences to predict the next move, and plays a counter. It tries 4-move patterns first, then 3-move patterns if none match, then 2-move patterns. More matching examples give a prediction more weight. If its history is still filling or no pattern matches, it uses Regular.

The two historical versions use the same basic approaches, with different rules for searching patterns or recording moves. The next page explains those changes.

What changed in v7.1 and v7.2

v7.1: Short-Term Memory became more selective

Before v7.1, Short-Term Memory recorded every opponent move, including moves from rounds it won. Since v7.1, it records only moves from rounds it tied or lost. It still keeps ten entries for each round position and counters the most frequent remembered move. Because winning rounds are skipped, those ten entries can cover a longer period of play.

Against the same five other bots, current Short-Term Memory won 6,123 games, compared with 5,484 for the old version, out of 25,000 games each. However, when the two versions played each other, the current version won 2,382 games and lost 2,600, with 18 ties. The update performed better against the shared opponents, but did not beat its predecessor directly.

v7.2: Pattern Detection checks more examples

Before v7.2, Pattern Detection tried 4-move patterns and then 3-move patterns if none matched. After finding a match, it skipped ahead by the pattern length. The current version moves ahead just one position at a time, so it also checks examples that share some moves. It adds a 2-move search when neither 4 nor 3 produces a match. Both versions remember fifteen opponent moves and use Regular when no pattern can be used.

The current bot still uses the first pattern length that produces matches: it does not combine 4-, 3-, and 2-move predictions. If the same prediction appears in several matches, it gets more weight in the random choice. Matches can overlap, so several examples can partly come from the same moves.

Against the same five other bots, current Pattern Detection won 14,176 games, compared with 12,568 for the old version, out of 25,000 games each. In their direct matchup, the current version won 2,027 games and lost 1,963, with 1,010 ties. Its overall improvement was larger than its small advantage in that direct matchup.

Each direct matchup combines 5,000 games across five seeds. These comparisons describe the simulated opponents in this report; results against people may differ.

How the opponent affects pattern success

Each cell shows the round win percentage, followed by how many rounds used that pattern length. The 4-move results are especially strong against the two Short-Term Memory versions.

Opponent	4-move: % / n	3-move: % / n	2-move: % / n	Fallback: % / n
Aggressive	39.94% / 10,990	38.14% / 8,592	35.74% / 4,882	33.84% / 461
Regular	37.78% / 10,320	36.27% / 9,328	35.72% / 5,011	31.58% / 266
Pattern Detection (pre-v7.2)	34.93% / 9,015	35.19% / 10,208	32.49% / 5,515	36.36% / 187
Random	34.24% / 9,120	33.11% / 10,635	33.41% / 5,067	40.78% / 103
Short-Term Memory	72.07% / 19,531	54.44% / 2,972	53.77% / 1,683	35.45% / 739
Short-Term Memory (pre-v7.1)	79.78% / 20,280	58.93% / 2,369	54.29% / 1,514	37.27% / 762

What if we leave out the two weakest opponents?

The next table uses only rounds against Regular, Aggressive, Random, and Pattern Detection (pre-v7.2). Leaving out both Short-Term Memory versions lowers the pattern win rates. This shows how much the opponent matters. The different pattern lengths still came from different situations.

Length	All six opponents win %	Other four win %	Other four loss %	Other four net %	Other four rounds
4-move	56.55%	36.91%	34.6%	+2.32%	39,445
3-move	38.06%	35.53%	33.6%	+1.95%	38,763
2-move	36.95%	34.29%	34.7%	-0.42%	20,475

Shorter sequences are easier to find by chance. Longer sequences are more specific, but appear less often. A high win rate can also come from an opponent that repeats easy-to-predict moves. The next page compares all three lengths on the same recorded rounds.

Compare pattern lengths on the same rounds

Whenever a 4-move pattern matched, shorter patterns could also have been used. We checked what each length could have predicted on those same rounds, using the opponent move that was actually played.

Length	All six: calculated win %	All six: calculated net %	Other four: calculated win %	Other four: calculated net %
4-move	56.58%	+29.58%	36.90%	+2.32%
3-move	55.50%	+27.37%	35.99%	+0.50%
2-move	54.43%	+25.36%	35.18%	-0.80%

All-six comparison: 79,256 recorded rounds. Other-four comparison: 39,445 rounds, excluding both short-term versions. Within each group, every pattern length is checked against the same recorded moves.

How the possible outcomes were calculated

Each matching example suggests a counter move. We check every suggestion against the recorded opponent move, then divide by that pattern length's number of suggestions. For example, if two of three suggestions would win, the chance of winning is two-thirds. Repeated suggestions count again, just as they do when the bot chooses. The percentages above average these chances across the recorded rounds.

What this comparison can tell us

These are calculated chances, rather than results from another tournament. They compare possible choices on the same past rounds. Actually changing the bot could change what both players learn and do next, so this does not tell us how a bot using only one pattern length would perform over whole games.

Does finding more examples help?

Selected length	One example: rounds / win %	Two examples: rounds / win %	3+ examples: rounds / win %
4-move	37,282 / 40.69%	12,237 / 46.69%	29,737 / 80.48%
3-move	23,686 / 38.35%	13,788 / 37.02%	6,630 / 39.22%
2-move	2,590 / 37.34%	4,034 / 37.85%	17,048 / 36.68%

Several matches may share some of the same moves, and an easy opponent may repeat the same sequence. More examples therefore do not guarantee a more reliable prediction.

Pattern results in each run and how we tested

Seed	4-move win %	3-move win %	2-move win %	Fallback win %	Pattern game W-L-T
42	56.51%	38.61%	36.15%	37.23%	3,274-1,742-984
43	56.56%	37.95%	38.07%	35.97%	3,248-1,782-970
44	58.10%	38.36%	35.83%	36.01%	3,251-1,816-933
45	56.28%	37.36%	36.87%	34.33%	3,236-1,802-962
46	55.25%	38.04%	37.82%	34.35%	3,194-1,804-1,002

Test design

Every bot plays every other bot, making 21 pairings. Each pair plays 1,000 games per run, with five rounds in every game. We repeat this with seeds 42-46: 105,000 games and 525,000 rounds altogether. Each bot plays 30,000 games. We rank bots by game wins, using fewer losses to break a tie.

Memory and randomness

Both bots start with empty memory when a new pairing begins in each run. They keep learning across its 1,000 games; their memory does not reset after every game. Both draw from the same repeatable random sequence, in a fixed order. We keep that order the same across all five runs.

Checking the results

We replayed all 105 saved pairing runs and matched every game and round total exactly. We also recorded how Pattern Detection chose its moves, without changing its choices. Its pattern, warmup, and fallback counts agree with the simulator and add up to all 150,000 rounds it played.

Interpreting the evidence

Bots keep learning between games, so later games depend on earlier ones. Five runs help show variation, but small differences may still be due to chance. The calculation on page 10 compares possible moves on recorded rounds; it does not replay a changed bot. These results apply to the rules and simulated opponents described here.

Every bot against its six opponents

All five runs combined, shown from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

1. Aggressive

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Pattern Detection	1,849	2,136	1,015	37.0%	8,869	9,590	6,541
Regular	2,423	1,905	672	48.5%	11,821	10,417	2,762
Pattern Detection (pre-v7.2)	1,912	2,106	982	38.2%	8,775	9,360	6,865
Random	1,971	1,966	1,063	39.4%	8,433	8,452	8,115
Short-Term Memory	4,060	779	161	81.2%	17,049	7,294	657
Short-Term Memory (pre-v7.1)	4,224	749	27	84.5%	17,435	7,474	91

2. Pattern Detection

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	2,136	1,849	1,015	42.7%	9,590	8,869	6,541
Regular	2,160	1,861	979	43.2%	9,177	8,463	7,360
Pattern Detection (pre-v7.2)	2,027	1,963	1,010	40.5%	8,622	8,656	7,722
Random	2,007	1,934	1,059	40.1%	8,398	8,210	8,392
Short-Term Memory	3,794	753	453	75.9%	16,882	5,856	2,262
Short-Term Memory (pre-v7.1)	4,079	586	335	81.6%	18,714	4,993	1,293

Every bot against its six opponents (continued)

All five runs combined, shown from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

3. Regular

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	1,905	2,423	672	38.1%	10,417	11,821	2,762
Pattern Detection	1,861	2,160	979	37.2%	8,463	9,177	7,360
Pattern Detection (pre-v7.2)	1,815	2,139	1,046	36.3%	8,389	9,090	7,521
Random	1,984	1,968	1,048	39.7%	8,531	8,385	8,084
Short-Term Memory	3,642	1,208	150	72.8%	15,667	8,749	584
Short-Term Memory (pre-v7.1)	3,863	1,107	30	77.3%	16,208	8,682	110

4. Pattern Detection (pre-v7.2)

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	2,106	1,912	982	42.1%	9,360	8,775	6,865
Pattern Detection	1,963	2,027	1,010	39.3%	8,656	8,622	7,722
Regular	2,139	1,815	1,046	42.8%	9,090	8,389	7,521
Random	2,059	1,971	970	41.2%	8,490	8,273	8,237
Short-Term Memory	2,824	1,376	800	56.5%	12,310	7,836	4,854
Short-Term Memory (pre-v7.1)	3,440	1,007	553	68.8%	15,006	7,036	2,958

Every bot against its six opponents (continued)

All five runs combined, shown from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

5. Random

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	1,966	1,971	1,063	39.3%	8,452	8,433	8,115
Pattern Detection	1,934	2,007	1,059	38.7%	8,210	8,398	8,392
Regular	1,968	1,984	1,048	39.4%	8,385	8,531	8,084
Pattern Detection (pre-v7.2)	1,971	2,059	970	39.4%	8,273	8,490	8,237
Short-Term Memory	1,983	2,007	1,010	39.7%	8,392	8,472	8,136
Short-Term Memory (pre-v7.1)	1,971	2,035	994	39.4%	8,475	8,629	7,896

6. Short-Term Memory

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	779	4,060	161	15.6%	7,294	17,049	657
Pattern Detection	753	3,794	453	15.1%	5,856	16,882	2,262
Regular	1,208	3,642	150	24.2%	8,749	15,667	584
Pattern Detection (pre-v7.2)	1,376	2,824	800	27.5%	7,836	12,310	4,854
Random	2,007	1,983	1,010	40.1%	8,472	8,392	8,136
Short-Term Memory (pre-v7.1)	2,382	2,600	18	47.6%	12,288	12,635	77

Every bot against its six opponents (continued)

All five runs combined, shown from each bot's own side. Each opponent pairing is 5,000 games and 25,000 rounds.

7. Short-Term Memory (pre-v7.1)

Opponent	Game W	Game L	Game T	Game win %	Round W	Round L	Round T
Aggressive	749	4,224	27	15.0%	7,474	17,435	91
Pattern Detection	586	4,079	335	11.7%	4,993	18,714	1,293
Regular	1,107	3,863	30	22.1%	8,682	16,208	110
Pattern Detection (pre-v7.2)	1,007	3,440	553	20.1%	7,036	15,006	2,958
Random	2,035	1,971	994	40.7%	8,629	8,475	7,896
Short-Term Memory	2,600	2,382	18	52.0%	12,635	12,288	77